



DR. B. C. ROY ENGINEERING COLLEGE, DURGAPUR

AN AUTONOMOUS INSTITUTE

AFFILIATED TO: MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST
BENGAL

Power Electronics

EE-504

B. Tech (EE) – 5th Semester (EE-2)

Comprehensive Course Lesson Plan & Syllabus

Prepared by:

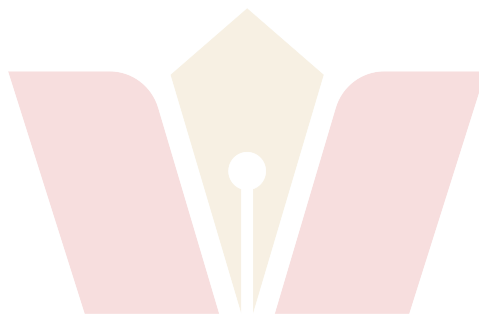
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1 Course Overview

This course introduces the principles, devices, and converter topologies that form the foundation of power electronics. Starting from the characteristics of power semiconductor switches, students are systematically taken through thyristor operation and triggering, phase-controlled AC-DC converters, DC-DC converters (choppers), DC-AC inverters, and resonant pulse converters, before concluding with real-world applications in motor drives, HVDC transmission, and static power quality equipment. The delivery sequence in this plan follows a device-progression order (self-commutated devices before line-commutated devices) rather than the strict syllabus module order, so that chopper circuits are covered immediately after the introductory module and before thyristor-based converters are introduced.

Course Fundamentals

 **Course Code:** EE-504

 **Credits:** 3

 **Contact Hours:** 4 hours/week

 **Section:** EE-2

 **Batch:** 2024–28

 **Semester:** 5th Semester (ODD)

 **Department:** Electrical Engineering

 **Institution:** Dr. B. C. Roy Engineering College, Durgapur

 **Contact Hours Note:** The approved syllabus lists 3 hours/week; the actual timetable allocation for this section is **4 hours/week**, and this plan is built on that basis.

1.1 Prerequisites

-  Network Theory
-  Analog Electronics
-  Electromagnetic Field Theory
-  Digital Electronics

2 Course Outcomes (CO)

Upon successful completion of this course, students will be able to:

Course Outcomes	Details / Statement	Action Verb	Knowledge Level
CO1	Define and explain the characteristics and operation of power electronic devices and switching components.	Define, Explain	Remember, Understand
CO2	Analyze and design different types of power electronic converters and their control circuits.	Analyze, Design	Analyze, Create
CO3	Evaluate the performance parameters of various power electronic converters and develop solutions for power factor improvement.	Evaluate, Develop	Evaluate, Create
CO4	Design and implement different control strategies for DC-DC converters and inverters.	Design, Implement	Create, Apply
CO5	Analyze and solve problems related to resonant pulse converters and their applications.	Analyze, Solve	Analyze, Apply
CO6	Apply power electronic concepts to real-world applications in motor control, HVDC transmission, and power systems.	Apply	Apply

Table 1: Course Outcomes for EE-504 Power Electronics

Module	CO1	CO2	CO3	CO4	CO5	CO6
Module 1: Introduction	3	–	–	–	–	–
Module 2: PNP Devices (Thyristors)	3	2	–	–	–	–
Module 3: Phase Controlled Converters	2	3	3	–	–	–
Module 4: DC-DC Converters (Chopper)	–	3	2	3	–	–
Module 5: Inverters	–	3	2	3	–	–
Module 6: Resonant Pulse Converters	–	2	–	2	3	–
Module 7: Applications	–	2	2	2	2	3

Table 2: CO to Module Mapping (3 = High, 2 = Medium, 1 = Low), as per approved syllabus

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	2	2	1	1	–	1	1	2
CO2	3	3	3	2	3	1	1	1	2	2	2
CO3	3	3	3	3	3	2	1	1	2	2	2
CO4	3	3	3	2	3	2	1	2	2	2	2
CO5	3	3	2	3	3	1	1	1	1	2	2
CO6	3	2	2	2	2	3	2	1	2	2	3

Table 3: CO to PO Mapping for EE-504

	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	1
CO2	3	3	1	2
CO3	3	2	2	2
CO4	3	3	1	2
CO5	3	2	1	2
CO6	3	3	3	2

Table 4: CO to PSO Mapping for EE-504

3 CO – Module and CO – PO – PSO Mapping

3.1 CO to Module Mapping

3.2 CO to PO Mapping

3.3 CO to PSO Mapping

4 Course Structure

i Syllabus Note: The course contains **7 Modules** with a total of **40 syllabus lectures** at **4 contact hours/week**. Module numbers follow the approved syllabus (Module 1 through Module 7); however, the **delivery order** in this plan is Module 1 → 4 → 2 → 3 → 5 → 6 → 7, so that self-commutated devices (Chopper) are covered before line-commutated devices (Thyristor, Phase Controlled Converters). All 40 lectures are accounted for; see the Week-wise Course Schedule for the exact chronology.

4.1 Module 1: Introduction (4 Lectures)

Learning Outcomes:

- Explain the scope and applications of power electronics.
- Distinguish ideal and practical characteristics of semiconductor switches.
- Identify losses in switching devices and their impact on converter design.
- Compare power diodes, power transistors, power MOSFETs, IGBT and GTO.

Lec. No.	Topic	Duration	Learning Activities
1.1	Concept of power electronics and applications of power electronic devices	50 min	 Interactive discussion
1.2	Ideal and practical characteristics of semiconductor switches; losses of switches	50 min	 Switching-loss curve study
1.3	Power diodes and power transistors: construction and switching behaviour	50 min	 Device datasheet study
1.4	Power MOSFET, IGBT and GTO: construction, characteristics and comparison	50 min	 Comparative device analysis

Key Topics – Module 1:

- Scope and industrial applications of power electronics
- Ideal vs. practical switch characteristics
- Conduction and switching losses
- Power diode, power transistor, power MOSFET, IGBT and GTO

4.2 Module 4: DC-DC Converters – Chopper (5 Lectures)

Learning Outcomes:

- Explain the principle of operation and control strategies of DC choppers.
- Classify chopper circuits by quadrant of operation.
- Evaluate chopper performance parameters.
- Describe multiphase chopper configurations.

Lec. No.	Topic	Duration	Learning Activities
4.1	Chopper: principle of operation, duty ratio and average output voltage	50 min	 Numerical workshop

Lec. No.	Topic	Duration	Learning Activities
4.2	Control strategies: time-ratio control (TRC) and current-limit control	50 min	⚙️ Control-scheme comparison
4.3	Step-up chopper: principle and derivation of output voltage expression	50 min	📈 Circuit and waveform analysis
4.4	Types of chopper circuits based on quadrant of operation (Class A to Class E)	50 min	📊 Quadrant classification exercise
4.5	Performance parameters of choppers; multiphase choppers	50 min	💻 eSim/LTspice simulation demo

🔑 Key Topics – Module 4:

- Duty ratio control and average output voltage/current
- Time-ratio control and current-limit control
- Step-up (boost) chopper operation
- One-quadrant to four-quadrant chopper classification
- Performance parameters and multiphase choppers

📌 **Module 4 Assessment:** Assignment-1: Chopper analysis and simulation (issued on completion of this module; due: **02-September-2026**, i.e. 2 weeks before CIA-01 submission).

4.3 📖 Module 2: PNP Devices – Thyristors (5 Lectures)

🚩 Learning Outcomes:

- Describe the thyristor family, symbols, V-I characteristics and applications.
- Explain the two-transistor model and turn-on methods of an SCR.
- Interpret switching and gate characteristics, and SCR ratings.
- Apply series/parallel operation and commutation techniques for SCR circuits.

Lec. No.	Topic	Duration	Learning Activities
2.1	Thyristor family: brief description of members, symbols, V-I characteristics and applications	50 min	🔍 Device family survey
2.2	Two-transistor model of SCR; SCR turn-on methods	50 min	🔧 Model derivation
2.3	SCR switching characteristics, gate characteristics and ratings	50 min	📈 Characteristic curve interpretation
2.4	SCR protection; series and parallel operation of SCRs	50 min	🔒 Protection circuit design

Lec. No.	Topic	Duration	Learning Activities
2.5	Gate triggering circuits; commutation techniques of SCR (natural and forced)	50 min	 Commutation-circuit problem solving

Key Topics – Module 2:

- Thyristor family members and V-I characteristics
- Two-transistor model and SCR turn-on methods
- Gate characteristics, ratings and SCR protection
- Series/parallel operation and gate triggering circuits
- Natural and forced commutation techniques

 **Module 2 Assessment:** Class Test-1 covering Module 1 (Introduction), Module 4 (Chopper) and Module 2 (Thyristors) (scheduled: **09-September-2026**, i.e. 1 week before CIA-01 submission). CIA-01 submission: **16-September-2026**.

4.4 Module 3: Phase Controlled Converters (6 Lectures)

Learning Outcomes:

- Analyze single-phase and three-phase phase-controlled converters with R, R-L and RLE loads.
- Evaluate the effect of free-wheeling diodes and source inductance on converter performance.
- Apply techniques of power factor improvement in phase-controlled converters.
- Explain single-phase and three-phase dual converters.

Part A: Single-Phase Converters (3 Lectures)

Lec. No.	Topic	Duration	Learning Activities
3.1	Single-phase half-wave and half-controlled converters with R, R-L and RLE loads	50 min	 Waveform derivation
3.2	Single-phase full-controlled converter with R, R-L and RLE loads	50 min	 Output voltage analysis
3.3	Effect of free-wheeling diodes and source inductance on converter performance	50 min	 Commutation-overlap discussion

Part B: Three-Phase Converters and Power Factor (3 Lectures)

Lec. No.	Topic	Duration	Learning Activities
3.4	Three-phase half-wave and half-controlled converters with R, R-L and RLE loads	50 min	 Circuit analysis
3.5	Three-phase full-controlled converters; external performance parameters	50 min	 Performance-parameter workshop
3.6	Techniques of power factor improvement; single-phase and three-phase dual converters	50 min	 Comparative case study

Key Topics – Module 3:

- Single-phase half-wave, half-controlled and full-controlled converters
- Three-phase half-wave, half-controlled and full-controlled converters
- Effect of free-wheeling diode and source inductance
- External performance parameters and power factor improvement
- Single-phase and three-phase dual converters

4.5 Module 5: Inverters (10 Lectures)

Learning Outcomes:

- Classify inverters by input source, output waveform, commutation and connections.
- Analyze single-phase and three-phase bridge inverters with R and R-L loads.
- Evaluate inverter performance parameters and PWM-based voltage control.
- Explain harmonic reduction strategies and the current source inverter (CSI).

Part A: Inverter Fundamentals and Bridge Circuits (5 Lectures)

Lec. No.	Topic	Duration	Learning Activities
5.1	Definition and classification of inverters: input source, output waveform, commutation, connections	50 min	 Conceptual discussion
5.2	Single-phase bridge inverter: principle of operation with R load	50 min	 Waveform sketching
5.3	Single-phase bridge inverter: principle of operation with R-L load	50 min	 Output analysis with inductive load
5.4	Three-phase bridge inverter: principle of operation with R load	50 min	 Six-step operation study
5.5	Three-phase bridge inverter: principle of operation with R-L load	50 min	 Line and phase voltage analysis

Part B: Performance, Modulation and CSI (5 Lectures)

Lec. No.	Topic	Duration	Learning Activities
5.6	Performance parameters of inverters: THD, distortion factor, harmonic factor	50 min	 Harmonic-spectrum interpretation
5.7	Voltage control using PWM technique – Part I: single-pulse and multiple-pulse modulation	50 min	 eSim/LTspice PWM demo
5.8	Voltage control using PWM technique – Part II: sinusoidal PWM	50 min	 eSim/LTspice SPWM demo
5.9	Harmonic reduction strategies in inverter output	50 min	 Filter-design discussion
5.10	Current source inverter (CSI): types, advantages, disadvantages and application areas	50 min	 VSI vs. CSI comparison

Key Topics – Module 5:

- Classification of inverters
- Single-phase and three-phase bridge inverters (R and R-L loads)
- Inverter performance parameters
- PWM-based voltage control and harmonic reduction
- Current source inverter (CSI)

 **Module 5 Assessment:** Assignment-2: Inverter and Phase Controlled Rectifier analysis and simulation (issued on completion of this module; due: **04-November-2026**, i.e. 2 weeks before CIA-02 submission).

4.6 Module 6: Resonant Pulse Converters (5 Lectures)

Learning Outcomes:

- Explain the need for resonant switching over hard switching.
- Analyze series and parallel resonant inverters.
- Distinguish zero-current switching (ZCS) and zero-voltage switching (ZVS) converters.
- Describe the resonant DC link inverter.

Lec. No.	Topic	Duration	Learning Activities
6.1	Introduction to resonant pulse converters; need for soft switching	50 min	 Hard vs. soft switching discussion
6.2	Series resonant inverter: principle and operation	50 min	 Resonant waveform analysis

Lec. No.	Topic	Duration	Learning Activities
6.3	Parallel resonant inverter: principle and operation	50 min	 Circuit topology study
6.4	Zero-current switching (ZCS) resonant converters	50 min	 Switching-loss reduction analysis
6.5	Zero-voltage switching (ZVS) resonant converter; two-quadrant ZVS; resonant DC link inverter	50 min	 eSim/LTspice soft-switching demo

Key Topics – Module 6:

- Series and parallel resonant inverters
- Zero-current switching (ZCS) resonant converters
- Zero-voltage switching (ZVS) resonant converter and two-quadrant ZVS
- Resonant DC link inverter

4.7 Module 7: Applications (5 Lectures)

Learning Outcomes:

- Apply power electronic converters to speed control of AC and DC motors.
- Explain the role of power electronics in HVDC transmission.
- Describe static circuit breakers and UPS systems.
- Explain the operation of a static VAR controller.

Lec. No.	Topic	Duration	Learning Activities
7.1	Speed control of DC motors using power electronic converters	50 min	 Drive-scheme discussion
7.2	Speed control of AC motors using power electronic converters	50 min	 VFD principle walkthrough
7.3	HVDC transmission: role of power electronic converters	50 min	 System-level overview
7.4	Static circuit breaker; uninterruptible power supply (UPS)	50 min	 Protection and backup systems study
7.5	Static VAR controller: principle and application	50 min	 Reactive power compensation discussion

Key Topics – Module 7:

- AC and DC motor speed control using power electronics
- HVDC transmission
- Static circuit breaker and UPS

- Static VAR controller

Module 7 Assessment: Class Test-2 covering Module 3 (Phase Controlled Converters), Module 5 (Inverters), Module 6 (Resonant Pulse Converters) and Module 7 (Applications) (scheduled: **11-November-2026**, i.e. 1 week before CIA-02 submission). CIA-02 submission: **18-November-2026**.

5 Assessment Framework

5.1 % Grading Distribution

Total Marks = End Semester Examination (60%) + CIA (40%)

CIA = Average of CIA-01 and CIA-02 (each out of 40 marks)

Component	Weight	Description
End Semester Examination	60%	Comprehensive theory examination
CIA (Continuous Internal Assessment)	40%	Average of CIA-01 and CIA-02

Table 14: Assessment Component Weights – EE-504

5.2 CIA Structure and Schedule

Each CIA is evaluated out of 40 marks with the following internal distribution:

CIA-01 Component	Marks	CIA-02 Component	Marks
Class Test 1	25	Class Test 2	25
Assignment 1	10	Assignment 2	10
Attendance	5	Attendance	5
Total	40	Total	40

Table 15: CIA Breakdown – EE-504

Important CIA Dates:

- Assignment-1 due:** 02-September-2026 (2 weeks before CIA-01 submission)
- Class Test 1:** 09-September-2026 (1 week before CIA-01 submission)
- CIA-01 Submission:** 16-September-2026
- Assignment-2 due:** 04-November-2026 (2 weeks before CIA-02 submission)
- Class Test 2:** 11-November-2026 (1 week before CIA-02 submission)
- CIA-02 Submission:** 18-November-2026

i CIA Calculation Rule: The final CIA marks are computed as the **average of CIA-01 and CIA-02**. For example, if CIA-01 = 34/40 and CIA-02 = 36/40, then CIA = $(34 + 36) / 2 = 35/40$ will be taken forward for the final grade.

5.3 🎓 Grading Scale

Grade	Range	Description
O	90–100	🏆 Outstanding
E	80–90	★ Excellent
A	70–80	👍 Very Good
B	60–70	✓ Good
C	50–60	— Average
D	40–50	✔ Pass
F	<40	✗ Fail

Table 16: Grade Distribution and Criteria

6 📅 Week-wise Course Schedule

i Schedule Note: 4 contact hours/week. Classes commence on **03-August-2026 (Monday)** and CIA-02 is submitted on **18-November-2026**, spanning **16 teaching weeks** before pre-examination activities begin on 23-November-2026. Lecture numbers follow the syllabus Module structure. *Rev* = Revision class; *CT* = Class Test. Total syllabus lectures = 40, all delivered by Week 13; Weeks 14–16 are reserved for revision, Class Test-2 and CIA-02 administration (this buffer also absorbs Durga Puja/Diwali holidays, exact dates to be confirmed against the final holiday list).

Week	Date (Monday)	Lectures	Topics Covered	Assessments / Remarks
1	03-Aug-2026	1.1, 1.2, 1.3, 1.4	Concept and applications of power electronics; ideal/practical switch characteristics; power diodes and transistors; MOSFET, IGBT and GTO	Course orientation
2	10-Aug-2026	4.1, 4.2, 4.3, 4.4	Chopper principle, duty ratio; TRC and current-limit control; step-up chopper; quadrant classification	—

Week	Date (Monday)	Lectures	Topics Covered	Assessments / Remarks
3	17-Aug-2026	4.5, 2.1, 2.2, 2.3	Chopper performance parameters, multiphase choppers (Module 4 complete); thyristor family, V-I characteristics; two-transistor model, turn-on methods	⚡ Assign-1 Issued
4	24-Aug-2026	2.4, 2.5, Rev, Rev	SCR protection, series/parallel operation; gate triggering, commutation techniques (Module 2 complete); revision	–
5	31-Aug-2026	Rev, Rev, Rev, Rev	Revision: Modules 1, 4, 2	⚡ Assign-1 Due: 02-Sep-2026
6	07-Sep-2026	CT-1, Rev, Rev, Rev	Class Test-1 (Modules 1, 4, 2); continued revision	✍ CT-1: 09-Sep-2026
7	14-Sep-2026	3.1, 3.2, 3.3, 3.4	Single-phase half/full-controlled converters; effect of free-wheeling diode; three-phase half/full-controlled converters	📋 CIA-01: 16-Sep-2026
8	21-Sep-2026	3.5, 3.6, 5.1, 5.2	Three-phase converter performance parameters, power factor improvement, dual converters (Module 3 complete); inverter classification; single-phase bridge inverter (R load)	–
9	28-Sep-2026	5.3, 5.4, 5.5, 5.6	Single-phase bridge inverter (R-L load); three-phase bridge inverter (R and R-L loads); inverter performance parameters	–
10	05-Oct-2026	5.7, 5.8, 5.9, 5.10	PWM voltage control (single/multiple pulse, sinusoidal); harmonic reduction strategies; current source inverter (Module 5 complete)	⚡ Assign-2 Issued
11	12-Oct-2026	6.1, 6.2, 6.3, 6.4	Need for soft switching; series and parallel resonant inverters; zero-current switching (ZCS) resonant converters	–
12	19-Oct-2026	6.5, 7.1, 7.2, 7.3	ZVS resonant converter, resonant DC link inverter (Module 6 complete); DC and AC motor speed control; HVDC transmission	–
13	26-Oct-2026	7.4, 7.5, Rev, Rev	Static circuit breaker, UPS; static VAR controller (Module 7 complete, full syllabus done); revision	Full syllabus complete

Week	Date (Monday)	Lectures	Topics Covered	Assessments / Remarks
14	02-Nov-2026	Rev, Rev, Rev, Rev	Revision: Modules 3, 5, 6, 7	Assign-2 Due: 04-Nov-2026
15	09-Nov-2026	CT-2, Rev, Rev, Rev	Class Test-2 (Modules 3, 5, 6, 7); continued revision	CT-2: 11-Nov-2026
16	16-Nov-2026	Rev, Rev, Rev, Rev	Final revision, doubt-clearing session	CIA-02: 18-Nov-2026

7 Assignments

 Software Note (Applicable to Option B – Software-based Assignment only): Only **open source / free** tools are to be used. Recommended platforms: **eSim** (open source, CDAC, esim.fossee.in) and **LTspice** (freeware, Analog Devices, analog.com/ltspice). No pirated or cracked software is permitted.

7.1 Assignment-1: Chopper (Module 4)

 Submission Note: Each student must submit **any one** of the two options below (Option A or Option B). Both options carry equal marks (10). All submissions must be **computer-typed only** (handwritten reports will not be accepted).

 Issue Date: Week 3 (17-Aug-2026) ** Due Date:** 02-September-2026

7.1.1 Option A – Classical Problem, Numerical and Theory

- **Topics:** Chopper duty ratio, average output voltage, quadrant classification, control strategies.
- **Questions (any five):**
 1. Derive the expression for average output voltage of a step-down chopper in terms of duty ratio α . A 220 V DC chopper feeds a resistive load of $10\ \Omega$ with a duty ratio of 0.6. Calculate the average and RMS output voltage, and the average output current.
 2. Explain time-ratio control (TRC) and current-limit control strategies for a chopper. Distinguish between constant-frequency and variable-frequency TRC with the help of waveforms.
 3. Derive the output voltage expression for a step-up chopper. A step-up chopper has input voltage 100 V and duty ratio 0.4. Calculate the output voltage and the average input current, if the output current is 5 A.
 4. Explain, with circuit diagrams, the classification of chopper circuits based on quadrant of operation (Class A to Class E). State the load and source conditions applicable in each case.
 5. A first-quadrant (Class A) chopper operates at a switching frequency of 1 kHz with a supply voltage of 200 V, feeding an R-L load with $R = 5\ \Omega$ and $L = 20\ \text{mH}$. For a duty ratio of 0.5, sketch the output voltage and current waveforms and calculate the peak-to-peak current ripple (assume continuous conduction).

6. What is a multiphase chopper? Explain its advantages over a single-phase chopper in terms of output ripple and filter requirements, with a neat circuit diagram.

- **Deliverables:** Complete typed solutions with all intermediate steps, neat diagrams, units and final answers clearly highlighted (computer-typed only, maximum 10 pages A4; figures MUST be drawn using QElectroTech or Inkscape etc. (free and open source, qelectrotech.org) is acceptable).

OR Complete it by hand and submit a scanned PDF.

7.1.2 Option B – Software-based Problem (eSim / LTspice)

- **Topics:** Buck (step-down), boost (step-up), and buck-boost chopper simulation.
- **Tasks:**
 - Build a buck converter circuit in eSim or LTspice with a given switching frequency and duty ratio; capture and label the output voltage and inductor current waveforms.
 - Vary the duty ratio (0.3, 0.5, 0.7) and tabulate the corresponding average output voltage from simulation; compare with the theoretical value $V_o = \alpha V_s$ and comment on the deviation.
 - Simulate a boost converter for a specified input voltage and duty ratio; plot the output voltage ripple and identify continuous vs. discontinuous conduction mode from the inductor current waveform.
- **Deliverables:** Circuit schematic (screenshot), simulation output waveforms (properly labelled), a comparison table of simulated vs. theoretical values, and a typed analysis note (computer-typed only, maximum 6 pages A4).

7.2 Assignment-2: Inverter and Phase Controlled Rectifier (Modules 5 & 3)

 Submission Note: Each student must submit **any one** of the two options below (Option A or Option B). Both options carry equal marks (10). All submissions must be **computer-typed only** (handwritten reports will not be accepted).

 Issue Date: Week 10 (05-Oct-2026) ** Due Date:** 04-November-2026

7.2.1 Option A – Classical Problem, Numerical and Theory

- **Topics:** Single-phase bridge inverter output analysis, phase-controlled rectifier firing angle control, performance parameters.
- **Questions (any five):**
 1. A single-phase full-bridge inverter is fed from a 220 V DC source and supplies a purely resistive load. Sketch the output voltage waveform and derive the RMS value of the fundamental component using Fourier analysis.
 2. A single-phase full-controlled bridge rectifier is fed from 230 V, 50 Hz supply and feeds an R-L load with a firing angle of $\alpha = 60^\circ$. Calculate the average and RMS output voltage, and comment on the effect of increasing α on the output.
 3. Explain the operating modes of a three-phase full-bridge (six-step) inverter with a neat circuit diagram and output voltage waveforms for a resistive load.

4. Define total harmonic distortion (THD) and distortion factor for inverter output. A single-phase inverter output contains a fundamental of 200 V (RMS) and a 3rd harmonic of 40 V (RMS). Calculate the THD (considering only the 3rd harmonic).
 5. Explain sinusoidal PWM (SPWM) technique for voltage control in a single-phase inverter. How does increasing the modulation index affect the fundamental output voltage and harmonic spectrum?
 6. A three-phase full-controlled converter feeds an R-L load from a 400 V, 50 Hz three-phase supply at a firing angle of $\alpha = 30^\circ$. Calculate the average DC output voltage and explain the effect of source inductance on commutation overlap.
- **Deliverables:** Complete typed solutions with all intermediate steps, neat waveform diagrams, units and final answers clearly highlighted (computer-typed only, maximum 10 pages A4; figures MUST be drawn using QElectroTech or Inkscape etc. (free and open source, qelectrotech.org) is acceptable).

7.2.2 Option B – Software-based Problem (eSim / LTspice)

- **Topics:** Single-phase full-bridge inverter and single-phase full-controlled rectifier simulation.
- **Tasks:**
 - Build a single-phase full-bridge inverter with resistive load in eSim or LTspice; capture the output voltage waveform and its FFT/harmonic spectrum.
 - Repeat with sinusoidal PWM switching and compare the fundamental RMS output voltage and THD against the square-wave (unmodulated) case.
 - Build a single-phase full-controlled rectifier with R-L load; vary the firing angle (30° , 60° , 90°) and tabulate the average output voltage from simulation against the theoretical expression.
- **Deliverables:** Circuit schematic (screenshot), labelled simulation waveforms, harmonic spectrum plot, a comparison table of simulated vs. theoretical values, and a typed analysis note (computer-typed only, maximum 6 pages A4).

8 Mini Project on Power Electronics (EE-504)

📌 Mini Project Note: Each group must submit **simulation and theory** for its allotted topic, using **open source / free software only** (LTspice, eSim, or equivalent). One topic per group; group members must jointly submit a single report and simulation files.

📅 Submission Deadline: 02-November-2026

8.1 Project Topics

1. Simulation and Theory of Flyback Converter
2. Simulation and Theory of Forward Converter
3. Simulation and Theory of Push-Pull Converter
4. SCR With and Without Snubber – Impact Analysis

8.2 👥 Group Allotment

Group	Topic	Roll Numbers	
Group 1	Flyback Converter	120016244052, 120016244054, 120016244056, 120016244058, 120016244060, 120016244062, 120016244064	120016244053, 120016244055, 120016244057, 120016244059, 120016244061,
Group 2	Forward Converter	120016244065, 120016244067, 120016253002, 120016253004, 120016253006, 120016253008, 120016253009	120016244066, 120016253001, 120016253003, 120016253005, 120016253007,
Group 3	Push-Pull Converter	120016253010, 120016253012, 120016253014, 120016253016, 120016253018, 120016253020, 120016253021	120016253011, 120016253013, 120016253015, 120016253017, 120016253019,
Group 4	SCR With and Without Snubber – Impact Analysis	120016253022, 120016253024, 120016253026, 120016253029, 120016253031, 120016253033	120016253023, 120016253025, 120016253028, 120016253030, 120016253032,

📌 Deliverables:

- Circuit schematic (LTspice/eSim), simulation waveforms with proper labelling
- Theoretical background: operating principle, key equations and design considerations
- Comparative discussion of simulated vs. theoretical results
- Typed report (computer-typed only), submitted as a single PDF per group

9 📖 Learning Resources

9.1 📖 Prescribed Textbooks

1. **Daniel W. Hart**, *Power Electronics*, McGraw-Hill Education.
2. **M. H. Rashid**, *Power Electronics*, PHI, 3rd Edition.
3. **P. S. Bimbhra**, *Power Electronics*, Khanna Publishers, 3rd Edition.

9.2 📖 Reference Books

4. **B. K. Bose**, *Modern Power Electronics & AC Drives*, Prentice Hall.
5. **Mohan, Undeland and Robbins**, *Power Electronics*, Wiley India.

6. **Phillip T. Krein**, *Elements of Power Electronics*, Oxford, 2007.
7. **J. P. Agarwal**, *Power Electronic Systems*, Pearson Education, 2006.
8. **M. S. Jamal Asghar**, *Power Electronics*, PHI, 2007.
9. **S. K. Pillai**, *Analysis of Thyristor Power Conditioned Motor*, University Press.
10. **J. M. Jacob**, *Power Electronics: Principles and Applications*, Thomson.

9.3 ★ Course Repository: GitHub

🏠 Open Access – EE-504: Power Electronics

🔒 Course slides and resources are hosted on an **open-access** GitHub repository. No sign-up required.

🔗 URL: <https://github.com/KingsukMajumdar/PowerElectronicsSlides>

📄 Students may view files directly on GitHub, or clone/download the full repository for offline access.

9.4 🌐 E-Learning Courses for Reference (From NPTEL/Others)

- **W1:** [NPTEL Lecture 1](#) – Dr. G. Narayanan, IISc Bangalore
- **W2:** [NPTEL Lecture 2](#) – Dr. G. Narayanan, IISc Bangalore
- **W3:** [Fourier Analysis of a Single PWM Signal & a Particular Harmonic Mitigation](#) – MATLAB File Exchange
- **W4:** [LPG Sensor Incorporating Microcontroller and Arduino Uno](#) – YouTube
- **W5:** [Impact of Switching Frequency on Converter DC-DC Buck Converter Parameters](#) – GitHub
- **W6:** [Duty Ratio Control Online Simulation](#) – NI Multisim Online

9.5 🖥️ Open Source / Free Software Tools

- **eSim** (esim.fossee.in): Open source EDA tool (CDAC/FOSSEE) for circuit simulation, including chopper, rectifier and inverter circuits.
- **LTspice** (analog.com/ltspice): Freeware SPICE simulator for power electronic converter analysis.
- **GNU Octave** (octave.org) and **Python 3** with NumPy, SciPy, Matplotlib: Numerical computation, waveform plotting and harmonic (FFT) analysis.
- **QElectroTech** (qelectrotech.org): Open source schematic capture for circuit diagrams.

10 Course Policies

10.1 Attendance Policy

- A minimum of **75%** attendance is mandatory as per institutional norms.
- Medical leave must be supported by a valid medical certificate submitted within 3 working days of rejoining.
- Students failing to meet the attendance requirement may be debarred from the end-semester examination.

10.2 Assignment Submission Policy

- Assignments must be submitted **on or before the stated due date**.
- A penalty of **10% of total marks per day** will be levied for late submissions.
- No assignment will be accepted beyond **one week** after the due date.
- Extension may be considered only for documented medical or institutional emergencies, subject to prior intimation.

10.3 Academic Integrity

- All submitted work must represent the student's own independent effort.
- Any form of plagiarism in assignments or examination papers will attract disciplinary action as per institutional rules.
- Peer discussion and collaborative learning are encouraged; however, each student must submit individually written reports and separate simulation files.
- Proper reference attribution is required wherever external sources are cited.

11 Faculty Contact Information

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 **Contact:** Available during office hours; appointments may be scheduled via email.
 <https://github.com/KingsukMajumdar>
 [Learn With Kingsuk](#)

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